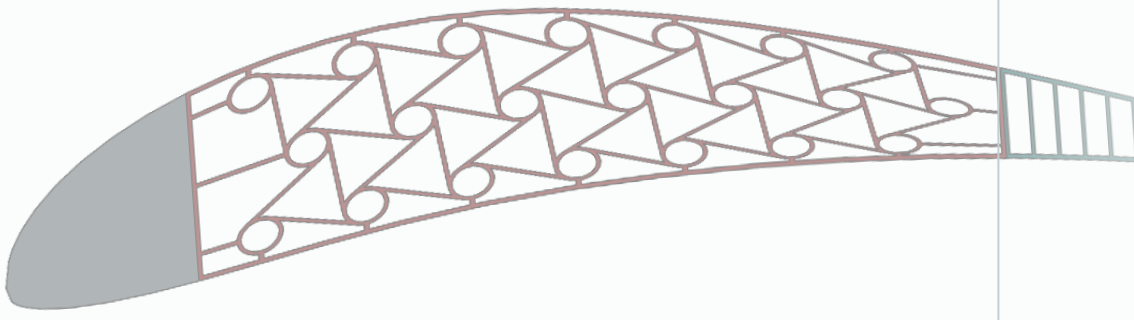
MECHANICS OF SOLIDS PROJECT

HEXACHIRAL HONEYCOMB

MECHANICS AND AN EPPLER 420 AIRFOIL APPLICATION ANALYTICAL DERIVATION, MATLAB CALCULATIONS AND THREE-DIMENSIONAL ANSYS STUDIES

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ABSTRACT

This project derives the in-plane properties of an ideal hexachiral honeycomb, evaluates the resulting equations in MATLAB form, and compares them with a finite three-dimensional compression specimen. The only coupon geometry reported has centre spacing R = 25 mm, in-plane ligament thickness t = 1 mm and ideal attachment radius r = 5 mm. Seven ANSYS load and mesh cases are reported. At the 1 mm reference stroke the simulation gives an apparent Poisson ratio of −0.8186 and an apparent Young modulus of 0.07525 MPa. The Spadoni–Ruzzene rigid-ring solution gives −0.9540 and 0.04527 MPa for the same analytical geometry and parent modulus. The mesh variations change apparent modulus by at most 1.24 % relative to the reference, so these mesh changes explain little of the discrepancy.

A separate application study maps a chiral core into a 330 mm chord Eppler 420 section and compares it with a straight-web core of nearly equal mass. At 0.50 mm of actuator stroke, the chiral tip travels 0.52890 mm compared with 0.41768 mm for the control. At 0.50 mm of matched tip travel the chiral design needs 21.0 % less actuator stroke and 2.53 times the mechanical work. The comparison quantifies a trade-off between stroke and actuator effort for the modelled airfoil segment.

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PART ONE

1

THE HEXACHIRAL CELL

Rotating rings, bending ligaments and a finite coupon

1 Project scope and evidence

The investigation has two scales. At the material scale, it asks how tangential ligaments and ring rotation produce in-plane auxeticity, and how closely an explicitly modelled finite lattice follows a periodic idealisation. At the section scale, it asks how a mapped chiral core changes the actuator stroke and work required to move an airfoil trailing edge.

Evidence	Scope and role
Analytical unit cell	Small-strain planar model with rigid rings and slender ligaments. Closed-form properties use ideal geometry.
3D coupon in ANSYS	Finite lattice, local X compression, TPU modulus input of 26 MPa. R01–R07 are the complete baseline series from [9].
Eppler 420 in ANSYS	Separate aluminium section models, fixed nose and vertical actuator stroke. Work and stress comparisons use [10].
Reduced airfoil model	Equivalent cantilever and rigid tail, followed by an illustrative thin-airfoil calculation.

Sections 4–7 derive the unit-cell response and check the kinematics, the thin-ligament limit and the published prefactors. The coupon comparison uses the workbook depth of 10 mm and parent modulus of 26 MPa throughout. The airfoil has its own 20 mm extrusion, aluminium parent material and boundary conditions. Its local mapped ligament geometry is different from the regular coupon.

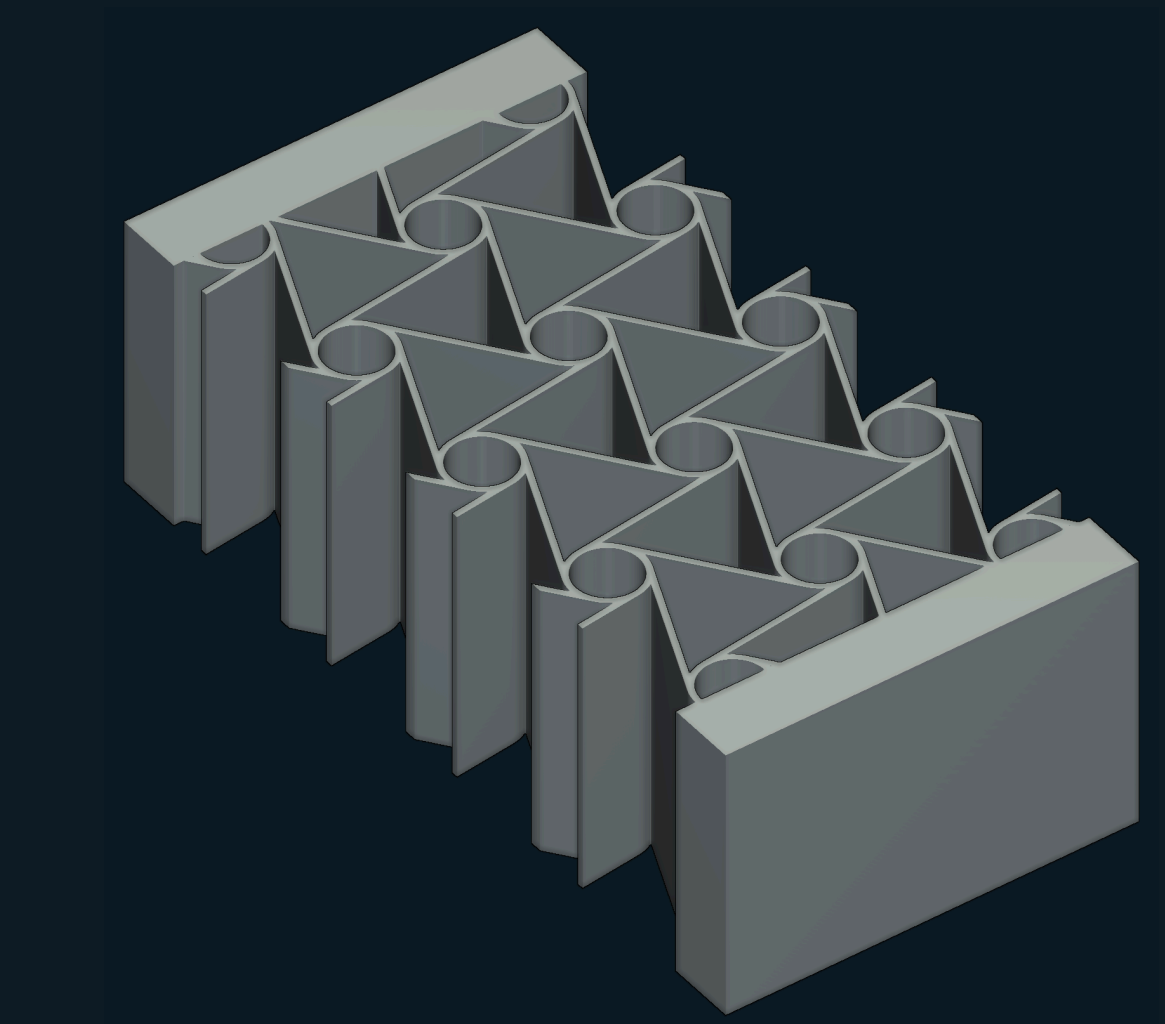
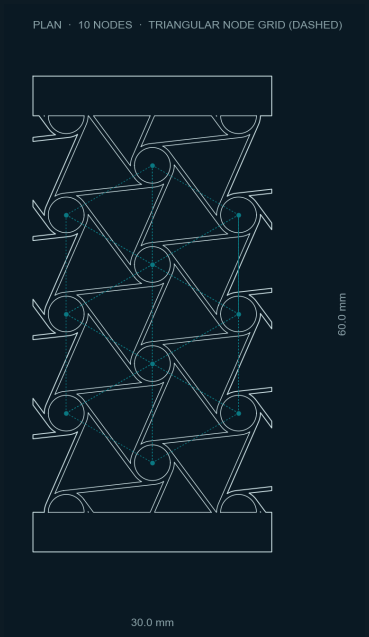


Fig. 0a The printed hexachiral sample, rendered from its CAD model (Chiral Honeycomb sample.STEP): ten rotating rings and their tangent ligaments between two solid loading plates, 20 mm deep.



The same sample in plan. Every ring sits on a triangular grid (dashed), and each ring carries ligaments that leave it tangentially — the geometry Sections 4–6 idealise as rigid nodes joined by bending beams. The two plates give the sample flat faces to load against.

Fig. 0b Plan of the printed sample with its overall dimensions and the node grid.

2 Notation

Symbol	Meaning	Unit
r	Ideal attachment lever arm to ligament centreline (map to CAD radii explicitly)	mm
R	Centre-to-centre spacing of neighbouring nodes	mm
L	Straight length of a ligament, between its two tangent points	mm
$t,\ d$	Ligament thickness (in plane) and depth (out of plane)	mm
β	Angle between a ligament and the line joining the node centres	degrees
E_s	Young's modulus of the solid material	MPa
I	Second moment of area of a ligament cross-section	mm⁴
A_n	Lattice area per node	mm²
ε	Dilation strain, equal along X1 and X2 (taken positive for contraction)	–
φ	Rotation of every node	rad
ψ	Rotation of the ligament chord relative to the line of centres	rad
θ	End rotation of a ligament relative to its own chord	rad
M	End moment on a ligament	N·mm
$E,\ K,\ \nu$	Effective in-plane Young's modulus, 2D bulk modulus, Poisson's ratio	MPa, MPa, –
t_n	Ring-wall thickness, separate from ligament thickness	mm

3 Assumptions

- **Rigid nodes.** Nodes rotate without deforming in the analytical model. A hollow printed ring meets this only approximately; record ring-wall thickness t_n separately and assess sensitivity to ring flexibility.
- **Slender bending ligaments.** Ligaments are Euler–Bernoulli beams. Sections 4–6 neglect axial compliance and transverse shear; Section 7.2 restores axial compliance. A small t/L supports this approximation, but t/L < 0.1 and a relative-density threshold are screening heuristics, not universal validity criteria for this topology.
- **Triangular lattice.** Node centres form a triangular lattice of spacing R. Each node carries six ligaments, each shared by two nodes, so there are 3 effective ligaments per node.
- **Symmetry and deformation mechanism.** Sixfold symmetry gives an isotropic classical in-plane response for the ideal lattice; it does not alone imply v = −1. Uniform dilation is the compliant mechanism when rigid nodes rotate and ligament axial extension is suppressed. The finite-compliance model in Section 7.2 tests this limiting interpretation.
- **Small strain, linear elastic material.** Kinematics and beam energy are used to first order in displacement or rotation and second order in energy. They are not a finite-strain model.
- **Bending-only limit.** Axial extension is constrained in Sections 4–6 rather than made energetically free. For the baseline, restoring axial compliance in Section 7.2 lowers the modulus slightly and shifts v above −1.

4 Geometry

In the ideal beam geometry, r is the perpendicular lever arm from the node centre to the ligament centreline, and L is measured between the ideal attachment points. The centreline tangent construction gives a right triangle with legs L and 2r and hypotenuse R. For CAD rings, state the outer and inner radii and map the attachment geometry to this idealisation; do not silently mix outer-surface and centreline dimensions.

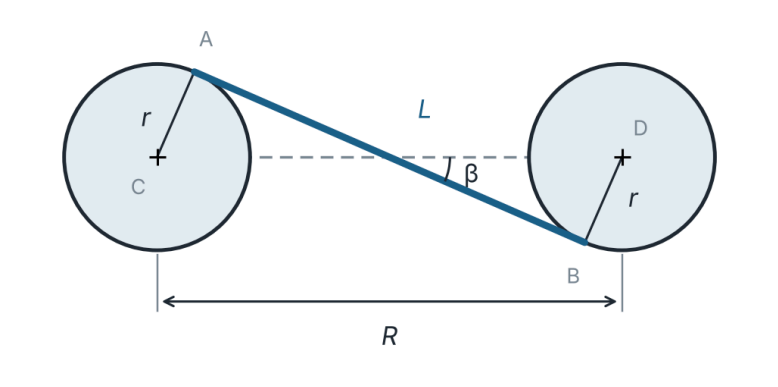


Fig. 1 Two neighbouring nodes C and D joined by ligament AB (undeformed). The ligament crosses the line of centres at angle β.

$$L = \sqrt{R^2 - 4r^2} \tag{1}$$

$$\sin\beta = \frac{2r}{R}, \qquad \cos\beta = \frac{L}{R} \tag{2}$$

$$I = \frac{d\,t^3}{12} \tag{3}$$

$$A_n = \frac{\sqrt{3}}{2} R^2 \tag{4}$$

Here A_n is the area of the lattice per node: the area of two equilateral triangles of side R, which matters later when counting ligaments per unit volume.

5 Kinematics and Poisson's ratio

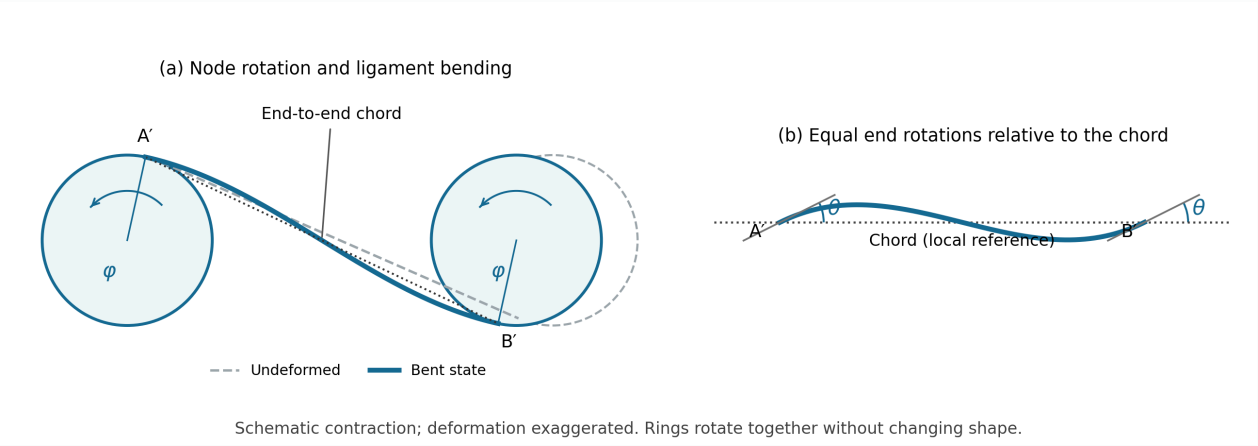


Fig. 2 Bent state of a ligament during ideal lattice contraction. (a) The nodes rotate in the same sense through φ and the ligament bends into an S shape; the dotted chord joins its displaced attachment points A′ and B′. (b) In a chord-aligned view, θ is the end-tangent rotation relative to the chord at both ends. The chord rotates clockwise by ψ in (a), giving θ = φ + ψ. Deformation exaggerated.

During the ideal dilation mechanism, nodes rotate together and ligaments bend into an S shape. Inextensibility preserves a ligament's centreline arc length, not its end-to-end chord length; chord shortening is higher order and is neglected. With θ the end rotation relative to the chord, the first-order separation relation is

$$D^2 \approx R^2 - 4rL\theta \tag{5}$$

$$\varepsilon = 1 - \frac{D}{R} \tag{6}$$

Using the contraction-positive strain of Eq. (6), linearisation gives

$$\varepsilon = \frac{2rL\theta}{R^2} \quad \Longleftrightarrow \quad \theta = \frac{\varepsilon R^2}{2rL} \tag{7}$$

The chord also turns relative to the line of centres, because β changes as the ends rotate. To first order,

$$\psi = \frac{2r}{L}\,\varepsilon \tag{8}$$

The end rotation relative to the chord is the sum of node and chord rotations, θ = φ + ψ. Substituting and using $R^2 - 4r^2 = L^2$ gives the node rotation in terms of the strain:

$$\varphi = \theta - \psi = \frac{L}{2r}\,\varepsilon \tag{9}$$

For the ideal uniform-dilation mechanism the two normal strains are equal, so the limiting uniaxial response is

$$\nu_{12} = \nu_{21} = -\frac{\varepsilon_2}{\varepsilon_1} = -1 \tag{10}$$

Equation (10) is a constrained, limiting response, not a general consequence of lattice symmetry. In an isotropic in-plane continuum $G = E/[2(1+\nu)]$; imposing ν = −1 at finite nonzero E is singular. Finite axial compliance gives a finite shear modulus and ν > −1 in the rigid-ring model. Edge effects, flexible rings and finite rotations can further change measured strain ratios.

One node, up close

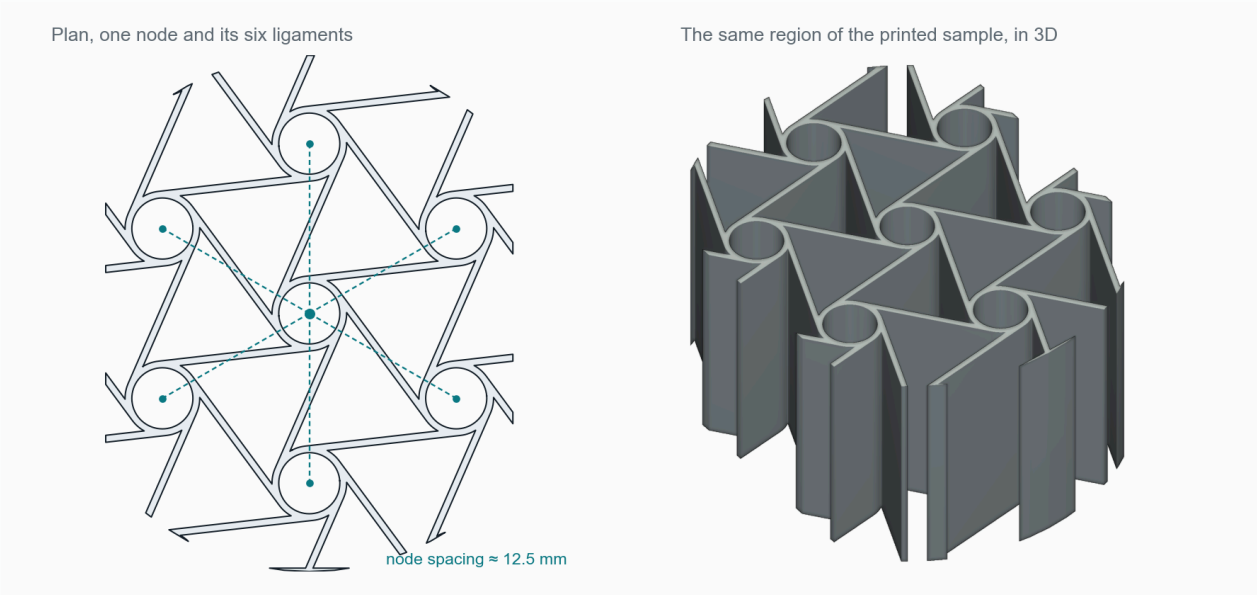


Fig. 2b One node of the printed sample and its six neighbours. Left: the plan, with the triangular node grid (dashed). Right: the same region in 3D. Every ligament leaves its ring tangentially, so rotating the rings winds the ligaments up rather than stretching them.

The mechanism in motion

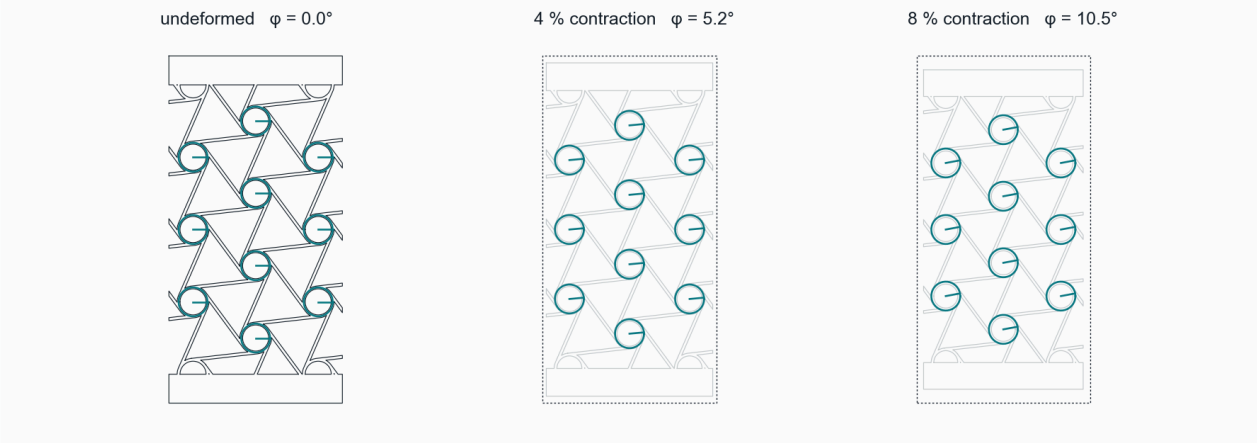


Fig. 2c Illustration of the ideal dilation mechanism of Section 5 applied to the sample's ring positions: as the lattice contracts by ε, every ring turns by the same φ = (L/2r) ε (Eq. 9), shown by the short ticks. The dashed box is the original outline. Schematic only — ring positions are scaled uniformly, not computed.

6 Ligament energy and Young's modulus

The derivation uses the cell-wall bending method of Gibson et al. [2]. With equal rotations θ at both ends relative to the chord, the ligament deforms into an antisymmetric S shape. The slope–deflection equations give the end moment and the stored energy:

$$M = \frac{6E_s I\theta}{L} \tag{11}$$

$$U_{\text{lig}} = M\theta = \frac{E_s\, d\, t^3\, \theta^2}{2L} \tag{12}$$

Each node has three effective ligaments, so the strain energy per unit volume is the energy of three ligaments divided by the node volume $A_n d$. Substituting θ from Eq. (7):

$$u = \frac{3U_{\text{lig}}}{A_n d} = \frac{\sqrt{3}\,E_s t^3\theta^2}{LR^2} = \frac{\sqrt{3}\,E_s t^3 R^2}{4r^2L^3}\,\varepsilon^2 \tag{13}$$

For equal normal strains the in-plane bulk modulus is defined by σ = 2Kε, giving u = 2Kε². In the limiting isotropic response ν → −1, E = 2K(1 − ν) → 4K, so the leading-order Young's modulus is

$$E = \frac{\sqrt{3}}{2}\,E_s \left(\frac{t}{L}\right)^3 \left(\frac{R}{r}\right)^2 \tag{14}$$

Using $R^2 = L^2 + 4r^2$, the geometric factor is $(R/r)^2 = (L/r)^2 + 4$. In the same limit the in-plane bulk modulus is

$$K = \frac{E}{4} \tag{15}$$

Equation (14) is cubic in t at fixed r and R and is the thin-ligament limit of Eq. (20) at fixed nonzero r/R. A first-order peak ligament bending-strain estimate, without joint concentrations, ring deformation, print defects or a failure criterion, is

$$\varepsilon_{\text{lig}} = \frac{3t\theta}{L} = \frac{3tR^2}{2rL^2}\,\varepsilon \tag{16}$$

7 Verification and comparison with the literature

7.1 Energy-method check

As an internal consistency check, an energy approach of the type used by Mousanezhad et al. [4] is applied to the ideal hexachiral cell — an adaptation, not an assertion that [4] publishes these equations verbatim. Under uniform equal-biaxial loading, half-turn symmetry eliminates the ligament midpoint moment, and rigid-node moment balance makes the bond force f act along the line of centres. Its transverse ligament component is f sin β. Summing the three bonds per node gives

$$\sigma = \frac{\sqrt{3}\, f}{R\, d} \tag{17}$$

The bending energy of the six half-ligaments (length L/2, load f sin β at the free end) is $U_{\mathrm{node}} = f^2 \sin^2\beta\, L^3/(8E_s I)$. Dividing by the node volume and equating it to σε gives

$$\frac{\varepsilon}{\sigma} = \frac{\sin^2\beta\, L^3}{\sqrt{3}\, E_s t^3} \tag{18}$$

$$K = \frac{\sigma}{2\varepsilon} = \frac{\sqrt{3}}{8}\, E_s \left(\frac{t}{L}\right)^3 \left(\frac{R}{r}\right)^2 \tag{19}$$

Using E = 4K reproduces Eq. (14). This is an alternative calculation within the same rigid-node, small-strain model, not independent experimental validation.

7.2 Closed form of Spadoni and Ruzzene

Spadoni and Ruzzene [3] assemble beam stiffness matrices, with axial and bending terms, for the hexachiral cell with rigid rings. With x = t/L, their closed forms (their Eqs. 45 and 46) are

$$\frac{E}{E_s} = \frac{4\sqrt{3}\,(1+x^2)}{2x^4\cos^2\beta + 2\sin^2\beta + 6x^2}\, x^3 \tag{20}$$

$$\nu = \frac{4x^2}{x^4\cos^2\beta + \sin^2\beta + 3x^2} - 1 \tag{21}$$

At fixed nonzero r/R (hence fixed nonzero sin β), taking x → 0 in Eq. (20) gives

$$\frac{E}{E_s} \to \frac{2\sqrt{3}}{\sin^2\beta}\, x^3 = \frac{\sqrt{3}}{2}\, x^3 \left(\frac{R}{r}\right)^2 \tag{22}$$

For finite x, Eq. (21) lies above −1. The shear modulus from [3], Eq. (48), is $G/E_s = (\sqrt{3}/4)(x + x^3)$, so G/E grows in the thin-ligament auxetic regime at fixed nonzero r/R even though both moduli tend to zero.

7.3 Comparison with Prall and Lakes

The printed Eq. (14) of Prall and Lakes [1] gives the following expression, transcribed without claiming every step of its derivation is consistent:

$$E_P = \sqrt{3}\, E_s \left(\frac{t}{L}\right)^3 \left(\frac{L}{r}\right)^2 \tag{23}$$

$$\frac{E}{E_P} = \frac{R^2}{2L^2} = \frac{1}{2\cos^2\beta} \tag{24}$$

At R/r = 5, Eq. (24) gives E/E_P = 25/42 ≈ 0.595: the printed Prall–Lakes expression is 1.68 times the present bending-only prediction. This compares ideal formulas; it does not prove that a physical specimen must follow either value.

Expression	E/E_s	Comparison	Interpretation
Present Eq. (14)	1.800×10^{-3}	Reference	Thin-ligament, rigid-node limit
Spadoni–Ruzzene Eq. (20)	1.741×10^{-3}	$E_{20}/E_{14} = 0.9673$	Finite axial compliance; rigid rings
Prall–Lakes Eq. (23)	3.024×10^{-3}	$E_P/E_{14} = 1.680$	Printed formula; inconsistency discussed below

Table 1. Verified formula comparisons at r = 5 mm, R = 25 mm, t = 1 mm.

A specific inconsistency is visible in [1], PDF page 4: the text counts 3/2 shared ribs but calls V = (√3/2)R²d a single triangular cell. A triangle of side R has area (√3/4)R²; the stated volume corresponds to two such triangles. At fixed volume and deformation, doubling the ligament count doubles the inferred modulus. Differences in strain–rotation kinematics must also be reconstructed consistently before assigning the whole prefactor discrepancy to individual causes.

7.4 Comparison with Bornengo, Scarpa and Remillat [5]

Bornengo et al. [5] restate the Prall and Lakes modulus in terms of β = t/r (note: [5] uses β for the thickness-to-radius ratio, not the ligament angle) and α = L/r. Since $\beta^3/\alpha = (t/L)^3(L/r)^2$, their Eq. (1) has the same dimensional scaling as Eq. (23):

$$E_B = E_s \,\frac{\sqrt{3}}{2}\,\frac{(t/r)^3}{L/r} = \frac{\sqrt{3}}{2}\, E_s \left(\frac{t}{L}\right)^3 \left(\frac{L}{r}\right)^2 \tag{25}$$

Equation (25) is half the printed Prall–Lakes expression; increasing the number of ligaments at fixed volume cannot explain a halving of stiffness. The ratio to the present bending-only expression is

$$\frac{E}{E_B} = \frac{R^2}{L^2} = \frac{1}{\cos^2\beta} \tag{26}$$

For R/r = 5, E/E_B = 25/21 ≈ 1.190; at t = 1 mm, $E_B/E_s \approx 1.512 \times 10^{-3}$, 16 % below Eq. (14). Separately, Bornengo's Table 1 does not reproduce its Eq. (1): at t/r = 0.05 and L/r = 4.6, 3.5 and 2.7, direct evaluation gives $10^5 E_B/E_s$ = 2.35, 3.09 and 4.01, whereas the table lists 1.18, 0.80 and 0.60 — disagreeing in magnitude and trend.

8 MATLAB function and baseline predictions

The attached chiral_models.m [11] is a MATLAB function file. Save it under that name, set the current folder to its location and call chiral_models(). Press Enter at the three prompts to use r = 5 mm, R = 25 mm and t = 1 mm. The function rejects R ≤ 2r, evaluates E/E_s and ν for the three models, prints a one-row comparison, and draws two analytical thickness sweeps at fixed r and R.

The numbers below were recalculated from the same expressions and checked against the workbook's Geometry models tab. The plots reproduce the sweep with 60 points from 0.30 to 3.00 mm and add one finite-element point. The thicker end of the curves should be read in light of the slender-beam assumption.

Model	E/E_s	E at E_s = 26 MPa (MPa)	ν
Project bending-only limit	0.0017998	0.046796	−1 (limit)
Spadoni–Ruzzene (SR)	0.0017410	0.045267	−0.954024
Prall–Lakes printed expression	0.0030237	0.078617	−1 (limit)
Bornengo et al. Eq. (1)	0.0015119	0.039308	—
ANSYS R01 finite coupon	0.0028944	0.075254	−0.818561

Table 2. Baseline analytical predictions and finite coupon result.

The ideal baseline has L = 22.91288 mm, β = 23.578°, L/R = 0.916515 and t/L = 0.043644. Per unit of contraction-positive dilation strain, the ring rotation is 2.2913 rad, the end rotation is 2.7277 rad and the peak ideal bending strain is 0.35714. For 1 % dilation these are about 1.31°, 1.56° and 0.357 %. The ideal bending result is independent of the common extrusion depth because d enters both ligament stiffness and homogenisation volume.

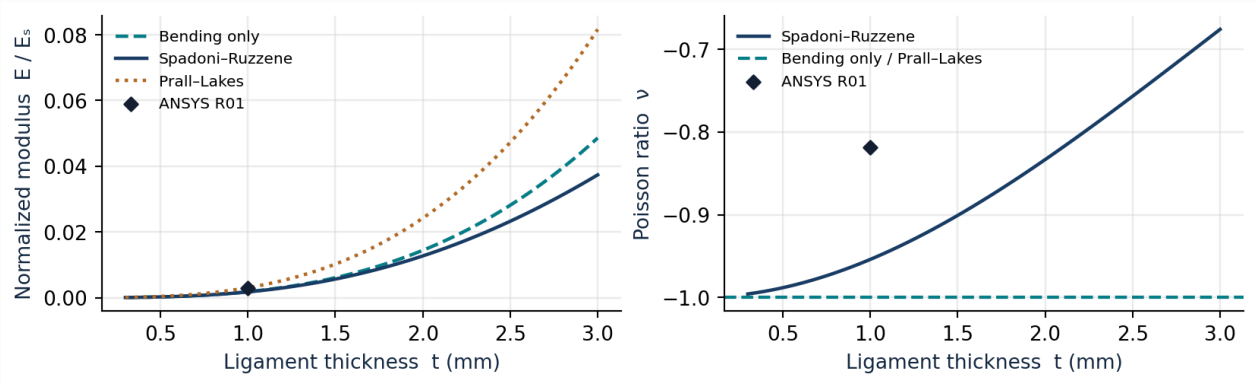


Fig. 3 Recalculated MATLAB analytical sweep and the single R01 finite coupon point. Bornengo's formula is discussed in Section 7.4 and is outside the three-model function.

9 Three-dimensional coupon simulation

9.1 Geometry, support and displacement probes

The coupon is an open-edge finite hexachiral patch. The specimen envelope is W = 136 mm perpendicular to the loading axis and d = 10 mm out of plane; the nominal gross section for stress normalisation is Wd = 1360 mm². The workbook uses r = 5 mm for the ideal calculation and records no separate ring wall, element order or global mesh size. The 3D solid rings are geometrically resolved. Each probe scopes two ring faces with a deformable MPC remote point, representing translations of the scoped rings without making the faces rigid.

In Ansys Mechanical the driven side has a prescribed local X translation with local Y and Z free; the opposing support is fixed, and the reaction is taken from it in the same local coordinates. The +1 mm X displacement drives the loaded side toward the support. Stroke is recorded as positive; gauge strains follow the tension-positive sign convention.

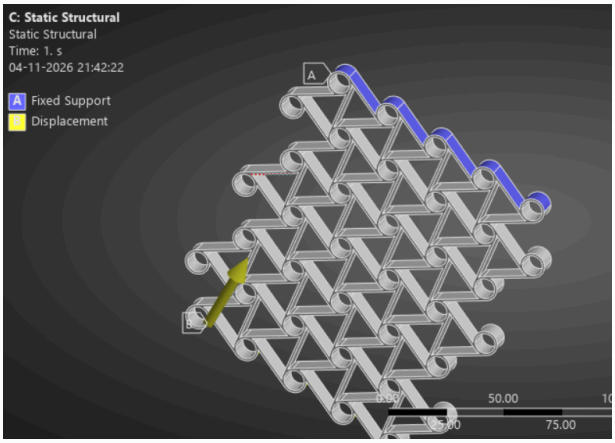


Fig. 4a Prescribed displacement and opposing support.

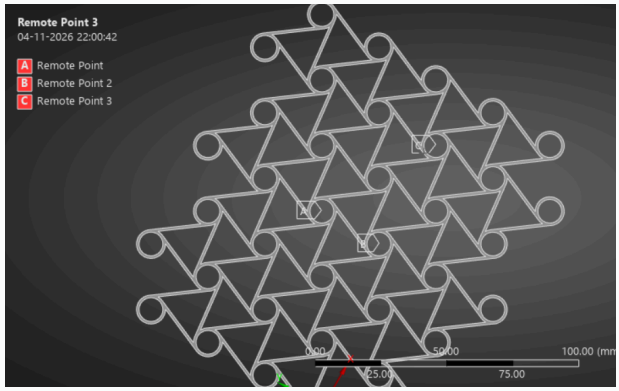


Fig. 4b Interior ring probes A, B and C. The gauges use B→C along local X and B→A along local Y.

The workbook uses initial signed separations $X_C - X_B$ = +43.30 mm and $Y_A - Y_B$ = +25.00 mm [9]. The complete initial coordinate triplets are absent from the results file, so exact alignment is a verification item.

$$\Delta u_x = u_x(C) - u_x(B), \qquad \Delta u_y = u_y(A) - u_y(B)$$

$$\varepsilon_x = \frac{\Delta u_x}{X_C - X_B}, \quad \varepsilon_y = \frac{\Delta u_y}{Y_A - Y_B}, \quad \nu_{\mathrm{app}} = -\frac{\varepsilon_y}{\varepsilon_x}$$

$$\sigma_n = \frac{|F_x|}{Wd}, \quad E_{\mathrm{app}} = \frac{\sigma_n}{|\varepsilon_x|}, \quad k = \frac{|F_x|}{q}$$

Here q is the imposed stroke magnitude. E_{app} combines gross-section stress with interior gauge strain; k uses the imposed stroke and measures force per unit of specimen-end motion. Both are secant quantities at the recorded stroke.

9.2 Recorded load and mesh cases

Run	Condition	Stroke (mm)	In plane (mm)	Sweep divisions	Depth step (mm)
R01	Baseline	1.00	0.33	4	2.50
R02	Half displacement	0.50	0.33	4	2.50
R03	Quarter displacement	0.25	0.33	4	2.50
R04	Double displacement	2.00	0.33	4	2.50
R05	8-division depth check	1.00	0.33	8	1.25
R06	Coarse in-plane mesh	1.00	0.50	4	2.50
R07	Fine in-plane mesh	1.00	0.25	4	2.50

Table 3. Baseline coupon load and mesh schedule. Depth spacing is nominal d divided by the sweep division count.

Run	F_x (N)	ε_x (%)	ε_y (%)	v apparent	σ_n (kPa)	E (MPa)	10^3 E/E_s
R01	−0.90163	−0.88097	−0.72113	−0.8186	0.66296	0.075254	2.8944
R02	−0.44986	−0.43937	−0.35119	−0.7993	0.33078	0.075284	2.8955
R03	−0.22603	−0.21938	−0.17314	−0.7892	0.16620	0.075759	2.9138
R04	−1.74430	−1.77208	−1.51883	−0.8571	1.28257	0.072377	2.7837
R05	−0.89106	−0.88098	−0.72114	−0.8186	0.65519	0.074371	2.8604
R06	−0.89832	−0.88109	−0.71925	−0.8163	0.66053	0.074967	2.8834
R07	−0.89033	−0.88085	−0.72139	−0.8190	0.65465	0.074321	2.8585

Table 4. Recalculated results from [9]. Forces retain the sign of the support reaction. Stress uses the gross section.

Reference calculation

For R01, Δu_x = 0.098560 − 0.480020 = −0.381460 mm and Δu_y = −0.039312 − 0.140970 = −0.180282 mm. The strains are −0.00880970 and −0.00721128. The support reaction is −0.901630 N; dividing its magnitude by 1360 mm² gives 0.000662963 MPa, and dividing by the axial strain magnitude gives E_{app} = 0.075254 MPa and E_{app}/E_s = 0.00289439. The common sign of the two strains confirms lateral contraction during axial compression.

Sensitivity to the imposed stroke

At fixed 0.33 mm in-plane sizing and four sweep divisions, v_{app} becomes more negative as stroke rises from 0.25 to 2.00 mm, from −0.7892 to −0.8571. Doubling the reference stroke nearly doubles axial gauge strain, while transverse contraction rises by about 2.11 times. The secant stiffness k falls from 0.90412 N/mm at 0.25 mm to 0.87215 N/mm at 2.00 mm, and E_{app} falls by 3.82 % from R01 to R04 — load dependence for this finite specimen, whose ring rotation, gauge location and end effects the dataset does not separate.

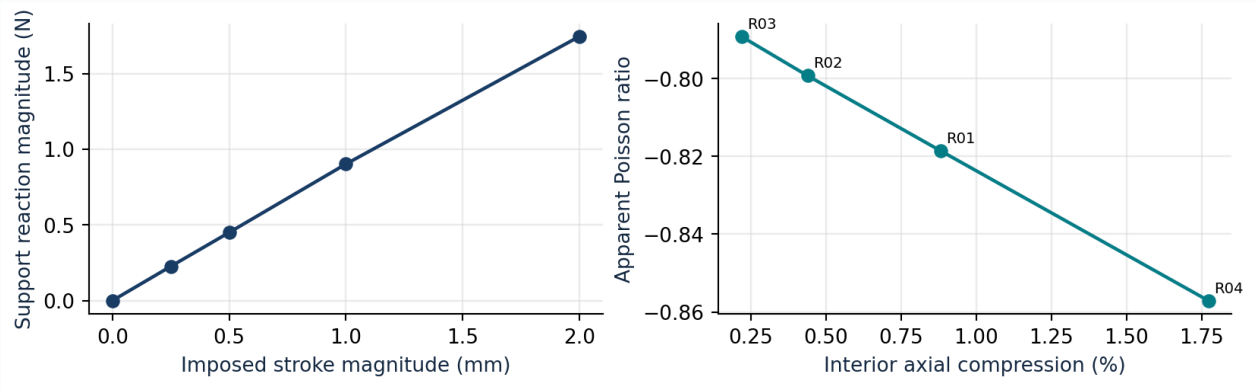


Fig. 5 Stroke sensitivity at fixed reference mesh. The left curve includes a zero-load reference; the right uses the axial strain measured by the interior gauge.

9.3 Mesh sensitivity and theory comparison

Change	Δv apparent	ΔE apparent (%)	Δ reaction magnitude (%)
R05 8-division depth check	−0.000012	−1.173	−1.172
R06 Coarse in-plane mesh	+0.002243	−0.381	−0.367
R07 Fine in-plane mesh	−0.000407	−1.240	−1.253

Table 5. Differences relative to R01 at the same 1 mm stroke.

The depth check and fine in-plane run reduce E_{app} by about 1.17 % and 1.24 %. Across the three mesh checks the largest change in v_{app} is 0.00225 and the largest reaction change is 1.25 %. The coarse–reference–fine values are not monotone in E_{app}, so these runs support a practical sensitivity estimate rather than a formal convergence order.

R01 is 0.1354 above the rigid-ring SR Poisson ratio and 66.2 % above its normalised modulus — gaps larger than the observed mesh effects. The analytical prediction assumes an ideal repeating lattice under a homogenised stress state; in the specimen, missing edge ligaments, end constraints, ring flexibility and short local gauges affect the comparison, as do the radius convention and gross width. The Prall–Lakes modulus is numerically closer to R01, but its idealised v = −1 and the different specimen conditions prevent that proximity from selecting a model.

A periodic unit-cell analysis with matched centreline geometry and parent properties would give the closest direct check of SR. In the finite coupon, repeating the strain extraction over more interior ring pairs would quantify gauge sensitivity.

PART TWO

2

EPPLER 420 AIRFOIL APPLICATION

A chiral core for a morphing trailing edge

10.1 Design and geometry mapping

The chiral mechanism is used here to distribute structural deformation within a continuous cambered section. Bornengo et al. [5] and the chiral airfoil study of Spadoni and Ruzzene [6] provide the application context. The project uses the normalised Eppler 420 ordinates (e420.dat) for the section profile; the 330 mm chord, 20 mm extrusion and internal partitions are project design choices.

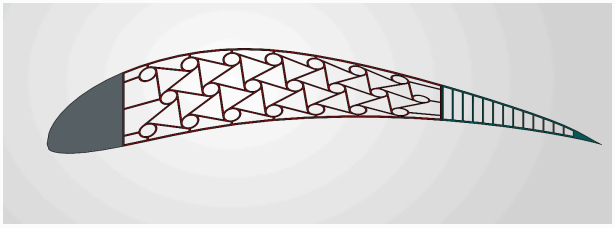


Fig. 6a Chiral middle core.

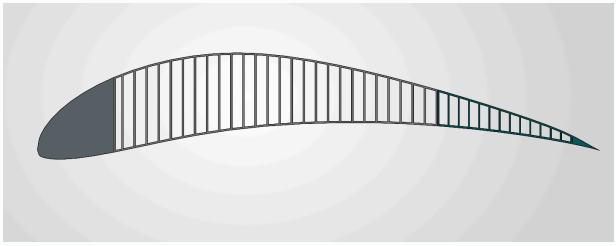


Fig. 6b Straight-web control. Both designs use a solid nose and a cellular tail.

Feature	Selected design
Section and partitions	Chord c = 330 mm, depth = 20 mm, nose to middle at a = 45 mm and middle to tail at m = 235 mm.
Middle and tail length	Middle length L_c = 190 mm. Tail length b = 95 mm.
Chiral reference lattice	21 rings in three staggered rows. Pitch 25 mm, mean ring radius 5 mm, ligament and ring wall 1 mm.
Outer and tail walls	Upper and lower walls 1 mm vertically. Chiral tail webs 0.8 mm chordwise at 6 mm pitch.
Compared core masses	Chiral 79.839 g and straight web 80.649 g, including the middle skin. Difference 0.810 g (1.0 %).

Let x increase aft from the nose, z upward, $z_u(x)$ and $z_l(x)$ be the profile ordinates, and ζ the vertical coordinate of the flat lattice relative to its mid-height, with reference height H_0 = 60 mm. The local thickness and initial camber are $h = z_u - z_l$ and $z_{c,0} = (z_u + z_l)/2$. The mapping, also described in the companion paper [12, Section 3], is

$$z(x, \zeta) = z_{c,0}(x) + \frac{\zeta}{H_0}\, h(x)$$

The mapping changes ring shape, local wall thickness and ligament orientation; a CAD mean ring radius is a separate definition from the ideal lever arm r. The matched control replaces the chiral middle lattice with straight webs using the same outer profile, support and actuator positions.

10.2 Reduced section and actuation models

For a first structural estimate, the solid nose is rigid, the middle a uniform Euler–Bernoulli cantilever of effective bending rigidity B (N·mm²), and the tail a rigid continuation of the junction tangent. The root is at x = a and the actuator gives downward stroke q at x = m. With $L_c = m - a$, $s = (x - a)/L_c$ and w the upward-signed displacement:

$$w(x) = \begin{cases} 0 & x < a \\ -\dfrac{q}{2}\,(3s^2 - s^3) & a \le x \le m \\ -q - \dfrac{3q}{2L_c}\,(x - m) & m < x \le c \end{cases}$$

The middle solution follows from a transverse end force on a clamped beam, with w(a) = w′(a) = 0, w(m) = −q and zero end moment. With b = c − m, the predicted tip travel, force and effective rigidity are

$$d_{te} = -w(c) = q\left[1 + \frac{3b}{2L_c}\right] = 1.75\, q$$

$$F_{\text{beam}} = \frac{3Bq}{L_c^3}, \qquad B = \frac{F L_c^3}{3q}, \qquad G_d = \frac{d_{te}}{q}, \qquad k_{\text{sec}} = \frac{F}{q}$$

For a monotonic quasistatic stroke, work is the integral of actuator force over stroke (1/1000 converts N·mm to J). For the discrete ANSYS data,

$$W_i = W_{i-1} + \frac{(F_{i-1} + F_i)(q_i - q_{i-1})}{2000} \quad [\text{J}]$$

For a target travel d* between tip readings d_1 and d_2, interpolate with $\lambda = (d^* - d_1)/(d_2 - d_1)$ for every recorded quantity:

$$q^* = q_1 + \lambda(q_2 - q_1), \quad F^* = F_1 + \lambda(F_2 - F_1), \quad W^* = W_1 + \frac{(F_1 + F^*)(q^* - q_1)}{2000}$$

Illustrative camber-to-lift estimate

For the thin-airfoil calculation [7], $x(\vartheta) = c(1 - \cos\vartheta)/2$ and $z_c = z_{c,0} + \Delta z_c$. At small slopes under two-dimensional attached flow, with angles in radians relative to the original chord,

$$\alpha_{L=0} = \frac{1}{\pi}\int_0^{\pi} z_c'[x(\vartheta)]\,(1-\cos\vartheta)\,d\vartheta, \qquad c_l = 2\pi(\alpha - \alpha_{L=0})$$

$$\Delta c_l = -2\int_0^{\pi} \Delta z_c'[x(\vartheta)]\,(1-\cos\vartheta)\,d\vartheta$$

Taking $\Delta z_c \approx w$ gives $\Delta c_l \approx 0.04468$ per mm of stroke: 0.02234 at q = 0.50 mm and 0.223 at an illustrative q = 5 mm, where the beam predicts 8.75 mm tip travel and 2.26° rotation. These are reduced-model calculations; the ANSYS ramp ends at 1 mm and provides no aerodynamic pressure or deformed-profile data for lift.

10.3 Airfoil finite element results

Both solid sections were analysed in Ansys Mechanical 2025 R1 with aluminium 6061-T6 throughout. The nose side faces were fixed, vertical motion was prescribed just behind the middle section, and vertical motion was measured at the trailing edge. The workbook reports default nose sizing and 0.8 mm nominal middle and tail sizing, with no separate mesh convergence series for the airfoil models.

Stroke (mm)	Chiral tip (mm)	Web tip (mm)	Chiral force (N)	Web force (N)	Chiral stress (MPa)	Web stress (MPa)
0.00	0.00000	0.00000	0.000	0.000	0.000	0.000
0.25	0.26492	0.20884	47.966	11.846	111.860	20.296
0.50	0.52890	0.41768	96.007	23.693	224.400	40.591
0.75	0.79193	0.62652	144.120	35.539	337.510	60.887
1.00	1.05410	0.83536	192.310	47.386	450.990	81.182

Table 7. Airfoil ANSYS readings from [10]. Travel and vertical reaction are magnitudes; stress is the peak equivalent stress in the middle section.

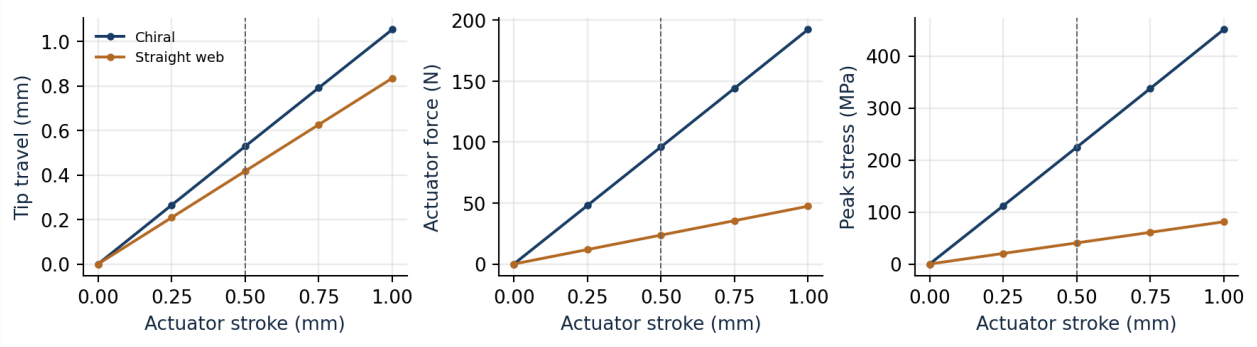


Fig. 7 Recorded tip travel, actuator force and peak stress for the two sections. Vertical guides mark the 0.50 mm comparison stroke.

At 0.50 mm of actuator stroke, the chiral tip travels 26.6 % farther and its actuator reaction is 4.05 times as high. The displacement gains are 1.058 and 0.835 mm/mm and the secant stiffnesses 192.0 and 47.4 N/mm. The chiral gain is 39.6 % below the reduced model's 1.75; tail flexibility, distributed actuation and the actual stiffness distribution can affect the gain.

Metric	Chiral	Straight web
Stroke (mm)	0.47263	0.59854
Actuator force (N)	90.75	28.36
Peak core stress (MPa)	212.1	48.6
Mechanical work (J)	0.02144	0.00849

Table 8. Interpolated comparison at a shared 0.50 mm trailing-edge travel.

At the shared target the chiral layout needs 21.0 % less stroke, 3.20 times the actuator force and 2.53 times the work. The 0.50 mm tip target lies within both ramps; a 1 mm target lies beyond the straight-web data. The interpolated peak stress compares reported maxima and does not track one material point.

10.4 Stress and aerodynamic interpretation

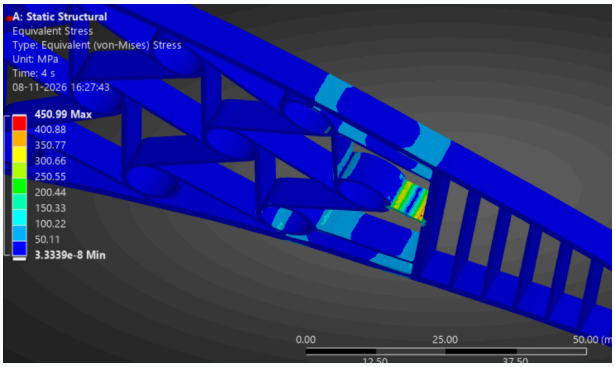


Fig. 8a Chiral core near the core–tail junction, 1.00 mm stroke.

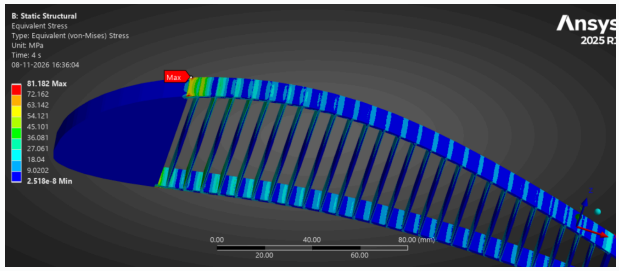


Fig. 8b Straight-web control near the upper nose–core transition. Each panel has its own colour scale.

The chiral core peaks at 337.51 MPa at 0.75 mm of stroke and 450.99 MPa at 1.00 mm, both above the nominal tensile yield strength of about 276 MPa for 6061-T6 [8]. Interpolating between 224.40 MPa at 0.50 mm and 337.51 MPa at 0.75 mm puts the 276 MPa crossing at about 0.614 mm of stroke. Both comparison points sit below yield: 224.40 MPa at 0.50 mm stroke and 212.08 MPa at 0.50 mm matched tip travel. The straight-web peak at 1.00 mm is 81.182 MPa.

This crossing is a screening estimate from reported peaks; the records identify the alloy but not the assigned yield strength or plasticity law. Confirm both in ANSYS Engineering Data and check mesh sensitivity at the stress concentration, then repeat the higher-stroke cases with an elastic–plastic law if an elastic hotspot exceeds yield.

The observed tip gains describe motion at one location. A camber comparison requires the full deformed upper and lower surfaces, interpolated onto a common chordwise grid; the change in their mean ordinate gives Δz_c, and smoothing that curve before differentiation supports the lift integral. Predicting pressure retention, pitching moment, separation, drag or flight performance would require additional loading or flow analysis. The companion paper [12] preserves the detailed CAD, mesh and stress images.

11 Project conclusions

The bending mechanism explains the negative Poisson ratio of the hexachiral lattice and gives a cubic stiffness dependence on ligament thickness in the slender limit. Adding axial compliance moves the ideal baseline prediction to −0.9540. The finite TPU coupon shows lateral contraction in all seven ANSYS baseline runs; its R01 apparent modulus and Poisson ratio differ from the ideal values by more than the mesh variations observed at the same stroke.

In the Eppler 420 section, the chiral core gives greater trailing-edge travel per unit of actuator stroke and requires higher force and work at the matched tip target. The chiral peaks at both 0.50 mm comparison points fall below the typical 6061-T6 yield value, while the 0.75 and 1.00 mm stroke peaks exceed it. The higher-stroke response therefore depends on confirming the material model and the mesh near the peak.

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[8] A. Mehmood et al. Experimental study to optimize cold working, aging temperature, and time on the properties of AA6061 tubes. *Frontiers in Materials* 10, 1199099 (2023).

[9] Hexachiral_Honeycomb_ANSYS_Results.xlsx. Coupon inputs and calculated results, R01–R07.

[10] Eppler420_Aerofoil_ANSYS_Comparison.xlsx. Simulation setup, recorded ramps, core masses and matched-travel calculations.

[11] chiral_models.m. MATLAB function for the three analytical lattice models.

[12] Uttej K. A chiral honeycomb core for a morphing Eppler 420 airfoil. Companion application paper.

A Coupon readings

The components below are the signed local translations from the Nodal inputs tab of [9]; the same workbook stores the axial reaction, applied stroke, initial separations and mesh settings.

Run	u_x(A)	u_y(A)	u_x(B)	u_y(B)	u_x(C)	u_y(C)
R01	0.49402	−0.039312	0.48002	0.14097	0.09856	0.02924
R02	0.24737	−0.018923	0.23997	0.068874	0.049721	0.014135
R03	0.12378	−0.0092751	0.11997	0.034009	0.024979	0.0069428
R04	0.98469	−0.087317	0.96104	0.29239	0.19373	0.06191
R05	0.49401	−0.039316	0.48002	0.14097	0.098557	0.029241
R06	0.49413	−0.039273	0.48027	0.14054	0.098757	0.02918
R07	0.49391	−0.039257	0.47994	0.14109	0.098532	0.029275

Table A1. Individual probe displacements in mm.

Run	Δu_x (mm)	Δu_y (mm)	k (N/mm)
R01	−0.3814600	−0.1802820	0.901630
R02	−0.1902490	−0.0877970	0.899720
R03	−0.0949910	−0.0432841	0.904120
R04	−0.7673100	−0.3797070	0.872150
R05	−0.3814630	−0.1802860	0.891060
R06	−0.3815130	−0.1798130	0.898320
R07	−0.3814080	−0.1803470	0.890330

Table A2. Recalculated pair displacements and force per unit actuator travel.

Traceability. Coupon probe readings occupy Nodal inputs rows 6–12; result formulas and values occupy Test results rows 7–13; the analytical baseline is on Geometry models row 7. In [10], the two ramp histories occupy rows 9–13 of Chiral and Straight Web, and the matched-travel outputs Comparison rows 19–22.

B MATLAB function

chiral_models.m [11]. The calculations are those used for Tables 1 and 2 and Fig. 3.

```
%% CHIRAL HONEYCOMB — three closed-form models from the literature
%
% Implements, for the hexachiral (tangent-ligament) honeycomb of
% Prall and Lakes, the three closed-form models compared in Table 2 of
% the project write-up:
%
%   1) Gibson & Ashby cell-wall-bending method, bending-only limit of
%      this derivation (Eq. 14). nu = -1 exactly.
%   2) Spadoni & Ruzzene (2012), rigid-ring closed form with both axial
%      and bending ligament deformation (their Eqs. 45-46, Eq. 20/21
%      of the note).
%   3) Prall & Lakes (1997), the original paper's own prefactor
%      (Eq. 23 of the note) — about 1.7x higher than (1) and (2) at
%      R/r = 5; this derivation and the literature check in Section 8
%      of the note settle that in favour of (1)/(2).
%
% Units: E is reported as the dimensionless ratio E/Es (set Es = 1
% throughout); multiply by your measured Es to get the physical
% modulus. r, R, t are in consistent length units (mm).
%
% This file is a function file (not a plain script) so that its local
% helper functions below are guaranteed to work the same way in any
% MATLAB version.
%
% Usage:
%   chiral_models()   % always prompts for r, R, t at the command line

function chiral_models()

clc;

%% ---- geometry: always prompt for it ----
r0 = input('Node radius r (mm) [default 5]: ');
if isempty(r0), r0 = 5; end
R0 = input('Node spacing R (mm) [default 25]: ');
if isempty(R0), R0 = 25; end
t0 = input('Ligament thickness t (mm) [default 1]: ');
if isempty(t0), t0 = 1; end

if R0 <= 2*r0
    error('Geometry is not valid: need R > 2*r (so that L = sqrt(R^2-4r^2) is real).');
end

fprintf('\nBaseline r=%.3g R=%.3g t=%.3g\n', r0, R0, t0);
[E14, nu14] = eq14(r0, R0, t0);
[E23, nu23] = eq23_prallLakes(r0, R0, t0);
[Esr, nusr] = eq20_spadoniRuzzene(r0, R0, t0);

fprintf('  Gibson & Ashby (bending-only)   E/Es = %.6e   nu = %.4f\n', E14, nu14);
fprintf('  Prall & Lakes (1997)            E/Es = %.6e   nu = %.4f\n', E23, nu23);
fprintf('  Spadoni & Ruzzene (2012)        E/Es = %.6e   nu = %.4f\n\n', Esr, nusr);

%% ---- Table 2 style comparison: only the geometry you entered ----
fprintf('%-14s %14s %16s %14s\n', 'r,R,t', 'Gibson-Ashby', 'Spadoni-Ruzzene', 'Prall-Lakes');
fprintf('%-14s %14.4e %16.4e %14.4e\n', ...
    sprintf('%g,%g,%g',r0,R0,t0), E14, Esr, E23);

%% ---- sweep t and plot all three E/Es curves (r, R fixed at input) ----
r = r0; Rfix = R0;
tvals = linspace(max(0.05, 0.3*t0), 3.0*t0, 60);
E14v = zeros(size(tvals)); Esrv = E14v; E23v = E14v;
nu14v = E14v; nusrv = E14v; nu23v = E14v;
```

```
for k = 1:numel(tvals)
    [E14v(k), nu14v(k)] = eq14(r, Rfix, tvals(k));
    [Esrv(k), nusrv(k)] = eq20_spadoniRuzzene(r, Rfix, tvals(k));
    [E23v(k), nu23v(k)] = eq23_prallLakes(r, Rfix, tvals(k));
end

figure('Name','E/Es vs t');
plot(tvals, E14v, '--', 'LineWidth', 1.5); hold on;
plot(tvals, Esrv, '-',  'LineWidth', 1.5);
plot(tvals, E23v, ':',  'LineWidth', 1.5);
xlabel('ligament thickness t (mm)'); ylabel('E / E_s');
legend('Gibson & Ashby (bending-only)','Spadoni & Ruzzene (2012)','Prall & Lakes (1997)','Location','northwest');
title(sprintf('r = %g mm, R = %g mm', r, Rfix)); grid on;
% --- overlay your own Ansys / physical-test points here, e.g.:
% ansys_t  = [1.0];
% ansys_E  = [ ... your measured E/Es values ... ];
% plot(ansys_t, ansys_E, 'ks', 'MarkerFaceColor','k', 'DisplayName','Ansys');

figure('Name','nu vs t');
plot(tvals, nu14v, '--', 'LineWidth', 1.5); hold on;
plot(tvals, nusrv, '-',  'LineWidth', 1.5);
plot(tvals, nu23v, ':',  'LineWidth', 1.5);
xlabel('ligament thickness t (mm)'); ylabel('\nu');
legend('Gibson & Ashby (bending-only)','Spadoni & Ruzzene (2012)','Prall & Lakes (1997)','Location','southeast');
title(sprintf('r = %g mm, R = %g mm', r, Rfix)); grid on;

end % chiral_models

%% ====================== local functions ======================

function [E, nu] = eq14(r, R, t)
% Gibson & Ashby cell-wall-bending method, bending-only closed form
% (this derivation), Eq. (14). nu is exactly -1.
    L = sqrt(R^2 - 4*r^2);
    E = (sqrt(3)/2) * (t/L)^3 * (R/r)^2;
    nu = -1;
end

function [E, nu] = eq23_prallLakes(r, R, t)
% Original prefactor of Prall & Lakes (1997), Eq. (23). Same t^3 scaling
% as Eq. (14) but about 1.7x higher at R/r = 5 (see Table 1 of the note).
    L = sqrt(R^2 - 4*r^2);
    E = sqrt(3) * (t/L)^3 * (L/r)^2;
    nu = -1;
end

function [E, nu] = eq20_spadoniRuzzene(r, R, t)
% Spadoni & Ruzzene (2012), rigid-ring closed form with axial + bending
% ligament deformation, their Eqs. 45-46 (Eq. (20)/(21) of the note).
    L = sqrt(R^2 - 4*r^2);
    x = t / L;
    c2 = (L^2) / (R^2);
    s2 = 1 - c2;
    E = (4*sqrt(3)*(1 + x^2)) / (2*x^4*c2 + 2*s2 + 6*x^2) * x^3;
    nu = (4*x^2) / (x^4*c2 + s2 + 3*x^2) - 1;
end
```